Midterm Review Quiz

50 minutes • 50 points • Through continuous distributions

Grading note. Award method credit after an earlier error; avoid repeated arithmetic deductions. Suggested times total 45 minutes, leaving 5 minutes to check work. **Instructions.** Answer all five problems and show the essential reasoning. No calculator is needed. Exact expressions, including factorials, exponentials, and values of the standard normal CDF Φ, are acceptable. You may quote standard distribution and moment formulas. Geometric variables count trials *including the first success*; exponential parameters are *rates*. Reserve the final 5 minutes for checking.

1. A random permutation

10 points · about 9 minutes

Four travelers' luggage tags are shuffled uniformly among all 4! assignments, one tag per traveler. Let I_i indicate that traveler i gets the correct tag, and let $X = I_1 + I_2 + I_3 + I_4$.

(a) Find $\mathbb{P}(X = 0)$ using inclusion–exclusion. Show your count or sum. (4 points)

Solution. Let $A_i = \{I_i = 1\}$. Fixing a specified set of k travelers leaves $(4-k)!$ assignments. Hence

$$\mathbb{P}(X=0) = \frac{4! - \binom{4}{1}3! + \binom{4}{2}2! - \binom{4}{3}1! + \binom{4}{4}0!}{4!} = \frac{24 - 24 + 12 - 4 + 1}{24} = \frac{3}{8}.$$

Credit: Uniform denominator: 1; intersection counts: 1; alternating sum: 1; evaluation: 1.

(b) Find $\mathbb{E}[X]$ by using the indicators. Does this calculation require the indicators to be independent? (2 points)

Solution. Each traveler has probability 1/4 of receiving the right tag, so $\mathbb{E}[X] = \sum_{i=1}^{4} \mathbb{E}[I_i] = 4(1/4) = 1$. Linearity needs no independence.

Credit: Indicator calculation: 1; no independence requirement: 1.

(c) For $i \neq j$, find $\mathbb{P}(I_i = I_j = 1)$ and decide whether I_i, I_j are independent. Then compute $\mathrm{Var}(X)$, keeping the cross terms. (4 points)

Solution. Fixing two tags leaves 2! permutations, so $\mathbb{E}[I_i I_j] = 2!/4! = 1/12$, different from $(1/4)^2 = 1/16$. Thus the indicators are dependent. Using $I_i^2 = I_i$ gives

$$\mathbb{E}[X^2] = 4 \cdot \frac{1}{4} + 2\binom{4}{2}\frac{1}{12} = 2, \qquad \mathrm{Var}(X) = 2 - 1^2 = 1.$$

Credit: Pair probability: 1; independence comparison: 1; second moment with cross terms: 1; variance: 1.

2. One hidden choice, two observations

10 points · about 8 minutes

A machine is selected once: machine A with probability $1/3$ and machine B with probability $2/3$. The selected machine produces two items. Conditional on the selected machine, defect indicators for the items are independent. Each item from A is defective with probability $1/4$; each from B is defective with probability $1/8$. Let D_1, D_2 be the events that the first and second items are defective.

(a) Find $\mathbb{P}(D_1)$. (3 points)

Solution. By total probability,

$$\mathbb{P}(D_1) = \frac{1}{3}\cdot\frac{1}{4} + \frac{2}{3}\cdot\frac{1}{8} = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}.$$

Credit: Two weighted branches: 2; total: 1.

(b) Given that the first item is defective, find the probability that machine A was selected. (3 points)

Solution. Bayes' rule gives

$$\mathbb{P}(A \mid D_1) = \frac{\mathbb{P}(D_1 \mid A)\mathbb{P}(A)}{\mathbb{P}(D_1)} = \frac{(1/4)(1/3)}{1/6} = \frac{1}{2}.$$

Credit: Correct Bayes numerator and denominator: 2; evaluation: 1.

(c) Find $\mathbb{P}(D_2 \mid D_1)$. Are D_1, D_2 independent without conditioning on the machine? Justify your answer numerically. (4 points)

Solution. Conditional independence within each machine and the updated weights give

$$\mathbb{P}(D_2 \mid D_1) = \frac{1}{2}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{1}{8} = \frac{3}{16}.$$

Unconditionally $\mathbb{P}(D_2) = 1/6$, by the same calculation as (a). Since $3/16 \neq 1/6$, the events are dependent. Equivalently $\mathbb{P}(D_1 \cap D_2) = (1/3)(1/4)^2 + (2/3)(1/8)^2 = 1/32$, which differs from $1/36$.

Credit: Conditional mixture or joint-event method: 2; value: 1; numerical independence comparison: 1.

3. Counts, waits, and a complete collection

10 points · about 9 minutes

Each pack contains one of three sticker types, independently and uniformly. Packs may duplicate previously seen types.

(a) Let N count type-1 stickers in the first four packs. State its distribution and parameters, and find $\mathbb{P}(N = 2)$ and $\mathbb{E}[N]$. (3 points)

Solution. $N \sim \text{Bin}(4, 1/3)$, with

$$\mathbb{P}(N = 2) = \binom{4}{2}(1/3)^2(2/3)^2 = \frac{8}{27}, \qquad \mathbb{E}[N] = \frac{4}{3}.$$

Credit: Family and parameters: 1; binomial probability: 1; mean: 1.

(b) Exactly two distinct types have been collected. Let G be the number of *additional* packs through the first pack containing the missing type. State its distribution and parameter, and find $\mathbb{E}[G]$ and $\mathbb{P}(G > 2)$. (3 points)

Solution. Each additional pack contains the missing type with probability $1/3$, so $G \sim \text{Geom}(1/3)$, on $1, 2, \ldots$. Thus $\mathbb{E}[G] = 3$ and $\mathbb{P}(G > 2) = (2/3)^2 = 4/9$.

Credit: Geometric model with parameter: 1; mean: 1; tail: 1.

(c) Starting with no stickers, let T be the number of packs through the first complete collection. Split T into the waits for the first, second, and third distinct types, and compute $\mathbb{E}[T]$. Explain why T itself is not one geometric random variable. (4 points)

Solution. Let G_k be the wait for a new type after exactly k types have been seen, for $k = 0, 1, 2$. The success probabilities are $1, 2/3, 1/3$ and the means are $1, 3/2, 3$. Since $T = G_0 + G_1 + G_2$, linearity gives

$$\mathbb{E}[T] = 1 + \frac{3}{2} + 3 = \frac{11}{2}.$$

The chance of obtaining a new type changes from stage to stage; also $T \geq 3$, whereas a geometric variable on $1, 2, \ldots$ has positive probability of being 1. Stage independence is not needed for this mean.

Credit: Stage probabilities: 1; means: 1; sum: 1; reason the full collection time is not geometric: 1.

4. Choose the law and use its parameters

10 points · about 9 minutes

(a) Calls arrive with Poisson counts: an interval of length t hours contains Poisson($3t$) calls, and disjoint intervals have independent counts. Let M count calls in the next 20 minutes. State its distribution and parameter, and find $\mathbb{P}(M \geq 2)$. (3 points)

Solution. Twenty minutes is 1/3 hour, so $M \sim \text{Poisson}(1)$. Then $\mathbb{P}(M \geq 2) = 1 - e^{-1} - e^{-1} = 1 - 2e^{-1}$.

Credit: Family and converted parameter: 1; complement: 1; value: 1.

(b) For the same call model, let W be the time in hours until the next call. Use the probability of zero calls to find $\mathbb{P}(W > t)$ for $t \geq 0$. Identify the distribution of W and find $\mathbb{P}(W > 2 \mid W > 1)$. (3 points)

Solution. $\mathbb{P}(W > t) = \mathbb{P}(\text{Poisson}(3t) = 0) = e^{-3t}$. Thus $W \sim \text{Exp(rate 3)}$, and $\mathbb{P}(W > 2 \mid W > 1) = e^{-6}/e^{-3} = e^{-3}$.

Credit: Zero-count argument and survival function: 1; family and rate: 1; conditional probability: 1.

(c) A separate task's duration Y is normal with mean 10 and *variance* 4. Express $\mathbb{P}(8 < Y < 13)$ using Φ. (2 points)

Solution. The standard deviation is 2, so $\mathbb{P}(8 < Y < 13) = \Phi(3/2) - \Phi(-1)$.

Credit: Standardization with standard deviation 2: 1; CDF difference with correct endpoints: 1.

(d) Let U be uniform on $[0, 4]$. Find $\mathbb{P}(U > 3 \mid U > 2)$ and compare it with $\mathbb{P}(U > 1)$. Does U have the memoryless property? (2 points)

Solution. $\mathbb{P}(U > 3 \mid U > 2) = (1/4)/(2/4) = 1/2$, whereas $\mathbb{P}(U > 1) = 3/4$. The unequal probabilities disprove memorylessness.

Credit: Conditional probability: 1; comparison and conclusion: 1.

5. From a density to a bound

10 points · about 10 minutes

A continuous random variable X has density

$$f(x) = \begin{cases} cx, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Determine c. (2 points)

Solution. Normalization requires $1 = \int_0^2 cx\,dx = 2c$, so $c = 1/2$. The resulting density is nonnegative.

Credit: Normalization integral: 1; value: 1.

(b) Find the CDF $F(x)$ for every real x, and use it to find $\mathbb{P}(1 < X < 2)$. (2 points)

Solution.

$$F(x) = \begin{cases} 0, & x \le 0, \\ x^2/4, & 0 < x < 2, \\ 1, & x \ge 2. \end{cases} \qquad \mathbb{P}(1 < X < 2) = 1 - F(1) = \frac{3}{4}.$$

Credit: Complete piecewise CDF: 1; interval probability: 1.

(c) Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\mathrm{Var}(X)$. (4 points)

Solution.

$$\mathbb{E}[X] = \int_0^2 x\frac{x}{2}\,dx = \frac{4}{3}, \qquad \mathbb{E}[X^2] = \int_0^2 x^2\frac{x}{2}\,dx = 2.$$

Therefore $\mathrm{Var}(X) = 2 - (4/3)^2 = 2/9$.

Credit: First moment: 1; second moment: 1; variance identity: 1; variance value: 1.

(d) Use Chebyshev's inequality to give an upper bound for $\mathbb{P}(|X - 4/3| \ge 2/3)$. State why the inequality applies. (2 points)

Solution. The variable has finite mean and variance (it is bounded). Chebyshev gives

$$\mathbb{P}(|X - 4/3| \ge 2/3) \le \frac{2/9}{(2/3)^2} = \frac{1}{2}.$$

Credit: Finite-variance hypothesis and correct inequality: 1; evaluated bound: 1. This is an upper bound, not an asserted exact probability.